MIL-M-16125C
21 February 1963
SUPERSEDING
MIL-M-0016125B(SHIPS)
6 September 1961
MIL-M-16125A
26 September 1951

MILITARY SPECIFICATION

METERS, ELECTRICAL, FREQUENCY

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. - This specification covers switchboard and portable electrical frequency meters for use in presenting visually the frequency of 60 or 400 cycle per second electric power system by a marked dial and pointer.

1.2 Classification. -

1.2.1 Type designation. - The type designation of meters shall be in the following form, and as specified (see 3.24 and 6.2):

Mr	49	W
Component (1.2.1.1)	Style (1.2.1.2)	Color scheme (1.2.1.3)

400	A	H
Center scale frequency (1.2.1.4)	Frequency range (1.2.1.5)	Special features (1.2.1.6)

1.2.1.1 Component. - Meters covered by this specification are designated by the two letter symbol "MR".

1.2.1.2 Style. - The style is designated by a two-digit number which signifies the following characteristics:

1.2.1.2.1 First digit. -

1.2.1.2.1.1 Switchboard meters. - The first digit designates the normal size and shape of flange on switchboard meters, as follows:

4 - 4-1/2 inch, rectangular flange
6 - 6 inch, rectangular flange

1.2.1.2.1.2 Portable meters. - The first digit designates the accuracy classification of portable meters, as follows:

1 - 1.5 percent

1.2.1.2.2 Second digit. -

1.2.1.2.2.1 Switchboard meters. - The second digit designates the type of mounting and the scale length in angular degrees, as follows:

8 - Flush mounting, 90 degree scale
9 - Flush mounting, 250 degree scale

1.2.1.2.2.2 Portable meters. - The second digit designates the voltage rating, as follows:

1 - 115 volts
2 - 230 volts
8 - 115 and 230 volts

1.2.1.3 Color scheme. - The color scheme of the dial background, markings and pointer is designated by a single letter, as follows:

B - Black dial background, white markings and pointer.
W - White dial background, black markings and pointer.

1.2.1.4 Center scale frequency. - The center scale frequency shall be designated as follows:

60 - For meters designed for a center scale frequency of 60 cycles per second.
400 - For meters designed for a center scale frequency of 400 cycles per second.

1.2.1.5 Frequency range. - The frequency range shall be designated by a single letter, as follows:

A - 59 to 61 cycles per second
B - 55 or 56 to 65 cycles per second
C - 390 to 410 cycles per second
D - 350 to 450 cycles per second

1.2.1.6 Special features. - A single letter shall designate special features, as follows:

H - High-impact-shock resistant.

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:20Z
Check the source to verify that this is the current version before use.

1.2.1.7 Examples of type designations. -

1.2.1.7.1 Switchboard meter. - The type designation MR49W60AH signifies:

(a) MR - Electrical frequency meter.
(b) 49 - 4 1/2 inch rectangular flange, switchboard meter, flush mounting, 250 degree scale.
(c) W - White dial background with black markings and black pointer.
(d) 60 - Center scale frequency of 60 cycles per second.
(e) A - Frequency range of 59 to 61 cycles per second.
(f) H - High-impact-shock resistant.

1.2.1.7.2 Portable meter. - The type designation MR11W400 C signifies:

(a) MR - Electrical frequency meter.
(b) 11 - 1.5 percent accuracy class portable meter with voltage rating of 115 volts.
(c) W - White dial background with black markings and black pointer.
(d) 400 - Center scale frequency of 400 cycles per second.
(e) C - Frequency range of 390-410 cycles per second.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-78 - Plastic Material, Laminated, Thermosetting (for Designation Plates).
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests for.
MIL-G-3787 - Glass, Laminated, Flat; (except Aircraft).
MIL-S-6872 - Soldering Process, General Specification for.
MIL-E-15090 - Enamel, Equipment Light-Gray (Formula No. 11).
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

AMERICAN STANDARDS ASSOCIATION
PUBLICATION C39.1

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York 16, New York.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Ave., at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Qualification. - Electrical frequency meter furnished under this specification shall be a product which has been tested, and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products lists.

3.2 Definitions. - Definitions which are listed in the American Standards Association Publication C39.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:20Z
Check the source to verify that this is the current version before use.

are applicable to the various technical terms which appear in this specification, except as set forth in 3.2.1.

3.2.1 Free Indication. - Meters shall be considered as indicating freely if the pointer of the meter under test moves simultaneously with the pointer of the standard meter with which it is being compared, when the electrical energy supplied to both meters is gradually increased from zero to an amount sufficient to produce end-scale deflection of the meter under test.

3.3 Material. - The material for each part shall be as specified herein. When a definite material is not specified, a material shall be used which is adequate for the purpose intended and for meeting the requirements herein. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

3.3.1 Plastic material. -

3.3.1.1 Molded thermosetting. - Molded thermosetting plastic material conforming to MIL-M-14 shall be used, and shall be of the proper type for its intended use.

3.3.1.2 Laminated thermosetting. - Laminated thermosetting plastic materials used for identification plates shall conform to MIL-P-78.

3.3.2 Windows. - Material used for meter windows shall conform to class II of MIL-G-3787.

3.3.3 Fume-emitting material. - Material that emits deleterious or toxic fumes at 85°C. or lower shall not be used.

3.3.4 Coatings and finishes. - Coatings and finishes which will melt, crack, chip, or scale as a result of tests specified in section 4 shall not be used.

3.4 Design and construction. -

3.4.1 Balancing. - The moving element shall be balanced by a means which will provide an easy method of rebalancing, such as threaded lock- or split-nuts, or a wire helix moving along a cross arm and counterweight arm. The use of solder, or an adhesive such as shellac, to hold the balance weights or to function as a part of the balance weight, will not be acceptable.

3.4.2 Magnets. - If laminated or multiple-piece magnets are used, the laminations or parts shall be securely fastened together so that they will not separate or loosen as a result of tests specified herein.

3.4.3 Threaded parts. - All threaded parts shall be in accordance with Handbook H28. Whenever practicable, all threads shall be in conformity with the coarse-thread series. The fine-thread series shall be used only for applications that might show a definite advantage through their use. Where a special diameter-pitch combination is required, the thread shall be in accordance with Handbook H28.

3.4.3.1 Bolts, nuts, and machine screws. - All bolts, nuts, and machine screws shall be of a corrosion-resistant material, or shall have a corrosion-resistant treatment.

3.4.3.2 Locking of screw-thread assemblies. - All screw-thread assemblies shall be made vibrationproof. When practicable, split-type lock washers or equivalent means shall be provided under all nuts.

3.5 Design and construction of component parts. -

3.5.1 Cases. - Meter cases shall be made of molded thermosetting plastic material, metal, or other material as specified herein. The cases shall be designed and constructed with close-fitting joints to minimize the entrance of dust and moisture. The dimensions of switchboard meter cases shall be as specified. (See 3.24.)

3.5.1.1 Portable meters. - Portable meters shall be contained in a substantial case of hardwood, or molded plastic material. The case shall be provided with durable supporting feet and a suitable carrying handle. The container shall be provided with a hinged cover and a snap catch. Where an auxiliary carrying case is supplied for a portable meter, it shall be acceptable to the bureau or agency concerned.

3.5.1.2 Finish. -

3.5.1.2.1 Switchboard. - The portion of the case exposed to view from the front of the panel shall have a light gray finish in accordance with formula 11, class 2 of MIL-E-15090. No nickel or other bright trimmings shall be used. Metal cases shall be rendered resistant to corrosion prior to the application of the final finish.

3.5.1.2.2 Portable. - All metal fittings and trimmings shall be protected against corrosion. Wooden cases shall have a durable semitransparent finish, such as varnish. Plastic cases shall have a durable finish on the exposed portion.

3.5.1.3 Certificate. - Portable meters shall be provided with a certificate covered with clear transparent plastic, and attached to the case in a conspicuous place. The certificate shall be signed by

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:20Z
Check the source to verify that this is the current version before use.

a person authorized by the manufacturer and shall contain the following minimum information:

(a) Manufacturer's name.
(b) Type or model.
(c) Serial number.
(d) Date of calibration.
(e) Temperature correction.
(f) Accuracy classification.
(g) Volt ampere burden.

3.5.1.4 Window. - Meters shall be provided with a window of glass, or other transparent material, free from discoloration, scratches, striae, electrostatic effect or other detrimental effects which may interfere in any way with normal reading of the meter in daylight. The window shall be secured by clamps, or by other suitable means. Cement alone is not considered sufficient for holding the window in place.

3.5.1.4.1 Scale visibility. -

3.5.1.4.1.1 Switchboard meters. - The scale and window shall be so designed that when the dial is vertical the entire scale of the meter can be easily read at any angle up to 45 degrees below the horizontal centerline on the vertical centerline, and 20 degrees to either side of the vertical centerline on the horizontal centerline, with general illumination coming from above and in front and at an angle 45 degrees from the horizontal centerline. Under these conditions, there shall be no objectionable shadow on the graduated scale.

3.5.1.4.1.2 Portable meters. - The scale and window shall be so designed that the window will extend a minimum of 1/8 inch beyond all scale markings.

3.5.2 Dial and pointer. -

3.5.2.1 Dial. - The dial shall be made of stiff material firmly secured to its mounting and readily removable. The dial shall be of such material and so secured as to insure freedom from warping, fading, or other deterioration during the tests specified herein. Paper dials will be satisfactory only when adequately bonded to and backed by a metal support and when they meet all tests specified herein without fading, discoloring, warping, or other deterioration.

3.5.2.1.1 Scale. -

3.5.2.1.1.1 Length. - The scale length shall be as specified. (See 3.24.)

3.5.2.1.1.2 Divisions. - The value of each scale division shall, whenever practicable, be either one, two, or five of the units measured or any decimal multiple or submultiple of these numbers. The angle subtended by a scale division shall be not less than 0.5 degree on portable meters, nor less than 1 degree on switchboard meters.

3.5.2.2 Pointer. - The meter shall have a pointer so formed as to permit accurate readings at the usual distance from which the meter is viewed, and approximate readings up to the distance of legibility of the scale markings. The pointer shall be of rigid construction and suitable for the intended application, and shall not become damaged or distorted when subjected to any of the tests specified herein.

3.5.2.2.1 Pointer clearance (parallax). - Where the meter is not provided with means for avoiding errors in reading due to parallax, the clearance between the pointer index and the graduated scale shall not exceed 0.10 inch, nor be less than 0.03 inch. Portable meters shall have a mirror on the dial near the scale and directly beneath the knife-edge portion of the pointer to reduce reading errors due to parallax.

3.5.3 Means of adjustment. -

3.5.3.1 Accuracy adjustment. - Meters shall be provided with practical shop means of adjustment to accuracy.

3.5.3.2 Zero adjuster. - Instruments having spring control shall have a zero adjuster accessible from the front of the case. It shall provide for a range of adjustment above and below the zero point of not less than 3 percent of the scale length and preferably shall be capable of rotating through 360 degrees without damage to any part of the instrument, and without becoming inoperative. If the zero adjuster is not designed for 360-degree rotation, it shall be capable of withstanding a torque of 10 ounce-inches without damage to the zero adjuster or to any part of the instrument. The zero adjuster shall have sufficient friction to prevent it from shaking loose and changing adjustment when subjected to tests specified herein.

3.5.3.3 Mechanical zero. - The mechanical zero point on the scale shall be identified by a zero marked at that point on the scale. The mechanical zero point shall not fall on a major division.

3.5.4 Terminals and connections. -

3.5.4.1 Portable meters. - Binding-post terminals shall be so constructed that they cannot turn or twist in their mounting. Terminals shall be insulated caps with no live parts exposed on cap nuts.

Source: https://assist.dla.mil -- Downloaded: 202640-18T08:20Z
Check the source to verify that this is the current version before use.

3.5.4.2 Switchboard meters.- Terminal studs shall be firmly anchored to the base of the meter in such a manner as to prevent turning or movement sufficient to cause damage to internal construction. Each terminal stud shall be furnished with the necessary nuts or screws, and washers.

3.5.4.2.1 Polarity.- On all d.c. switchboard meters the left-hand terminal (looking at the rear of the meter in the normal mounting position) shall be the positive terminal. (See 3.22.1.2.)

3.5.4.2.2 Bonding.- All permanent-magnet, moving-coil meters shall have the magnetic system and dial electrically connected to one terminal. Insulated cases having metallic bezel rings shall have this ring connected to the same terminal by means of a brush contact so that the mechanism can be readily withdrawn.

3.5.4.2.3 Connection diagrams.- Meters having more than one circuit shall have a print showing the wiring diagram of connections, together with instructions attached permanently or sufficiently secured to prevent loss during shipment.

3.5.5 Mounting switchboard meters.- Switchboard meters shall be mounted by means of mounting bolts or studs. The necessary mounting bolts (screws), nuts, and lock washers shall be supplied with each meter.

3.5.5.1 Four-and-one-half-inch meters.- The 4 1/2 inch meters shall mount on a panel drilled in accordance with the specified panel-drilling plan. (See 3.24.)

3.5.5.2 Six-inch meters.- The 6-inch meters shall mount on a panel drilled in accordance with the specified panel-drilling plan (see 3.24), or shall be furnished with a suitable adapter to permit mounting without drilling additional holes.

3.6 Calibration.-

3.6.1 Position.- Meters shall be calibrated in the position in which they are normally used, and shall meet the specified accuracy requirements in such position at the reference temperature (25°C).

3.6.2 Mounting panel.- Switchboard meters shall meet the specified accuracy requirements when mounted on a nonmagnetic panel and also on a magnetic-steel panel 1/8 inch thick.

3.7 Position influence (balance).- When switchboard meters are tilted 60 degrees from their normal mounting position, in any direction, the pointer tip shall not deviate from the zero mark more than the amount specified. (See 3.24 and 4.3.3.)

3.8 Initial accuracy.- At the reference temperature, the initial accuracy, expressed as a percentage of the actual scale range, shall be within the limits specified. (See 3.24 and 4.3.4.)

3.9 Damping factor.- When tested in accordance with 4.3.5, the damping factor shall be not less than the value specified. (See 3.24.)

3.10 Response time.- When tested in accordance with 4.3.6, the response time shall not exceed the value specified. (See 3.24.)

3.11 Power consumption.- The power consumption shall not exceed the value specified. (See 3.24 and 4.3.7.)

3.12 Frequency range.- Meters shall cover one of the following ranges, as specified:

55 or 56 to 65 cycles.
59 to 61 cycles.
350 to 450 cycles.
390 to 410 cycles.

3.13 External-field influence.- Switchboard meters and portable meters shall have sufficient magnetic shielding so that the effect of an external magnetic field of 5 oersteds will not cause a change in indication greater than the value specified. (See 3.24 and 4.3.8.)

3.14 Heat effect at 65°C.- Meters shall indicate freely (see 3.2.1) during and after the test specified in 4.3.9. The heat effect at 65°C. (the difference between corresponding readings of steps 9 and 11 of table III) and the permanent change in indication (the difference between corresponding readings of steps 1 and 13 of table III) shall not exceed the values specified. (See 3.24.)

3.15 Temperature influence.- The temperature influence shall not exceed the value specified. (See 3.24 and 4.3.10.)

3.16 Low-temperature exposure.- During exposure to a temperature of 0°C., meters shall indicate freely. (See 3.2.1.) The change in indication at the low temperature, and the permanent change in indication, shall not exceed the values specified. (See 3.24 and 4.3.11.)

3.17 Dielectric stress.- When tested in accordance with 4.3.12, meters shall withstand a voltage of 2,600 volts alternating current without damage or flashover.

3.18 Insulation resistance.- The insulation resistance shall be not less than 20 megohms. (See 4.3.13.)

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:20Z
Check the source to verify that this is the current version before use.

3.19 Vibration. - When meters are tested in accordance with 4.3.14, no screws, bearings, pivots, or other parts shall be so loosened or damaged as to cause a change in indication greater than the value specified. (See 3.24.)

3.19.1 Friction effect. - After completion of the vibration test specified in 4.3.14, the friction effect shall not exceed the value specified. (See 3.24 and 4.3.14.1.)

3.20 Humidity resistance. - After the test specified in 4.3.15, meters shall operate freely, and the permanent change in indication shall not exceed the value specified. (See 3.24.)

3.21 Shock (high-impact). - When high-impact-shock resistant meters are tested in accordance with 4.3.16, no screws, bearings, pivots, windows, or other parts shall become loosened, cracked, or unduly damaged. Cracking of glass windows, not to exceed 4 cracks which extend to the length of the window, is permissible. The cracks shall not impair reading of the meter scale. The permanent change in indication based on calibration made after resetting the pointer to zero or null position (if applicable) shall not exceed the value specified. (See 3.24.)

Note. - The design of high-impact-shock-resistant meters shall be such that disassembly and proper reassembly prior to the shock test will not impair the shock resistance of the meter.

3.22 Marking. - Meters shall be permanently and legibly marked with the applicable information specified in 3.22.1 and 3.22.2.

3.22.1 Terminal identification. -

3.22.1.1 Portable meters. - The terminals of multiple voltage meters shall be identified by the proper voltage rating on or near the applicable terminal.

3.22.1.2 Switchboard meters. - Polarity identification of d.c. meter terminals shall be near the positive terminal. (See 3.5.4.2.1.)

3.22.2 Dials. - Dial markings shall be sharply defined and visible from the front of the case. These markings shall not, however, be so conspicuous as to distract attention from the scale markings. Unless otherwise specified (see 3.24), the scale markings and numerals, and items (a) and (c) below, shall be marked on the dial; items (b), (d), (e), and (f) shall be marked on the dial on the front of the meter or on an attached identification plate conforming to MIL-STD-130.

(a) Designation of quantity measured.
(b) Manufacturer's type or model number, or both.
(c) Manufacturer's name or trade-mark (or symbol) or both.
(d) Type designation (see 1.2.1).
(e) Serial number.
(f) External accessories as required.
Example: Use external impeder
Serial 1234 rated 120V.

3.23 Workmanship. - Meters shall be manufactured and processed in a careful and workmanlike manner in accordance with good design and sound practice. The interiors of meters shall be free from metal filings, grease or oil, foreign material, dust or other loose particles which might interfere with the normal operation of the meters.

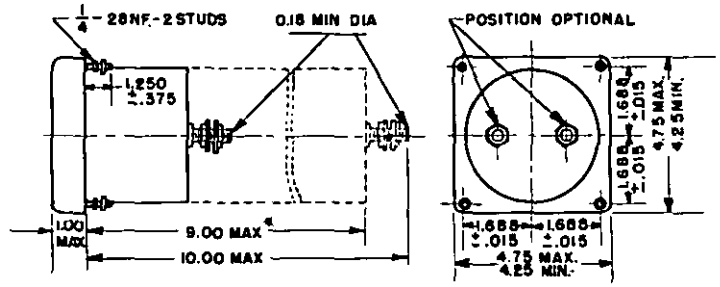
3.23.1 Soldering. - Soldering shall be in accordance with MIL-S-6872.

3.24 Detail requirements for individual meter types. - Detail requirements applicable to individual meter types are specified in 3.24.1 to 3.24.4 inclusive. In the event of any conflict between the general requirements of this specification, and the detail requirements, the latter shall govern. (See 6.2.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:20Z
Check the source to verify that this is the current version before use.

3.24.1 Detail requirements:

250 Degree Nominal Scale, 4-1/2 Inch Rectangular
Switchboard Frequency Meters



*THIS DIMENSION HAS BEEN CHOSEN TO ALLOW THE MOUNTING OF ACCESSORIES ON INSTRUMENT PROPER.

ALL DIMENSIONS IN INCHES

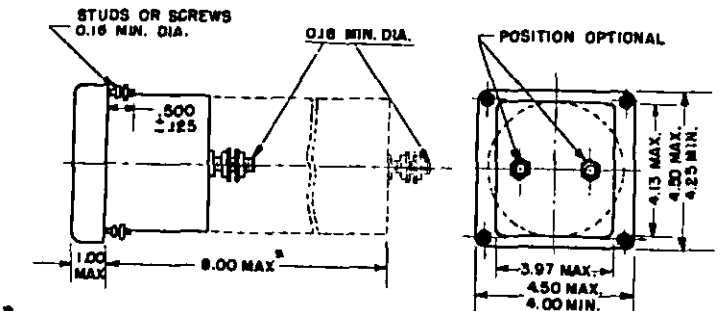
Sh 5897

ITEM	REQUIREMENT
1. Scale length	5 inches min.
2. Normal operating position	Mounted on vertical panel.
3. Position influence	60 degree tilt from normal operating position, 1 percent of scale length max. with rated voltage applied.
4. Rated accuracy	3.0 percent of frequency span.
5. Damping factor	1.5 min. (overshoot, 67 percent max.).
6. Response time	5 seconds max. at rated voltage.
7. Loss	6 va max. at middle frequency with rated voltage applied.
8. External temperature influence	1 percent of frequency span max. at middle frequency with ±10°C variation from +25°C with rated voltage applied.
9. External field influence	4 percent of frequency span max. at middle frequency in 5 oersted field with rated voltage applied.
10. Safe working voltage to ground	1100 volts max. peak value.
11. Dielectric test	Live parts to face and panel: 2600 volts a. c.
12. Max. voltage	20 percent greater than rated value.
13. Voltage influence	1.5 percent of frequency span with ±10 percent voltage variation; 4 times this max. error is allowed for a ±20 percent variation.
14. Temperature, permanent effect	2.0 percent of frequency span.
15. Vibration, permanent effect	2.0 percent of frequency span.
16. Humidity	3.0 percent of frequency span.
17. Class HI shock, permanent effect	7.5 percent of frequency span based on initial calibration.

Source: https://assist.dla.mil -- Downloaded: 202640-18T08:20Z
Check the source to verify that this is the current version before use.

3.24.2 Detail requirements:

90 Degree Scale, 4-1/2 Inch Rectangular Switchboard Frequency Meters



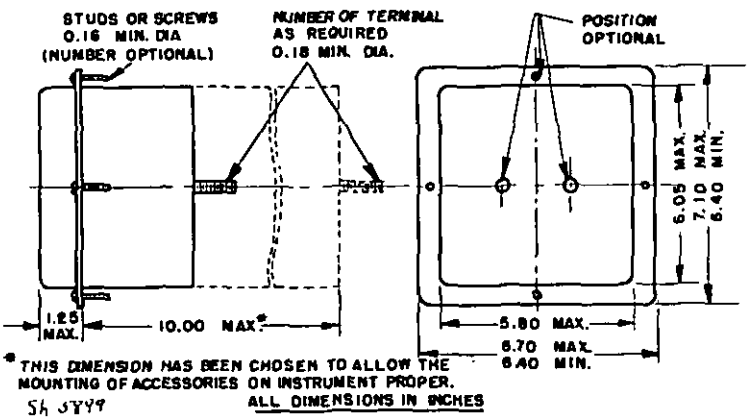
Sh 5899

ITEM	REQUIREMENT
1. Scale length	3.2 inches min.
2. Normal operating position	Mounted on vertical panel.
3. Position influence	60 degree tilt from normal operating position, 1 percent of scale length max. with rated voltage applied.
4. Rated accuracy	3.0 percent of frequency span.
5. Damping factor	1.5 min. (overshoot, 67 percent max.)
6. Response time	5 seconds max. at rated voltage.
7. Loss	5 va max. at middle frequency with rated voltage applied.
8. External temperature influence	1 percent of frequency span max. at middle frequency with ±10°C. variation from +25°C. with rated voltage applied.
9. External field influence	4 percent of frequency span max. at middle frequency in 5 oersted field with rated voltage applied.
10. Safe working voltage to ground	1100 volts max. peak value.
11. Dielectric test	Live parts to face and panel: 2600 volts a.c.
12. Max. voltage	20 percent greater than rated value.
13. Voltage influence	0.5 percent of frequency span with ±10 percent voltage variation; 4 times this max. error is allowed for a ±20 percent variation.
14. Temperature, permanent effect	2.0 percent of frequency span.
15. Vibration, permanent effect	2.0 percent of frequency span.
16. Humidity	3.0 percent of frequency span.

Source: https://assist.dla.mil -- Downloaded: 202640-18T08:20Z
Check the source to verify that this is the current version before use.

3.24.3 Detail requirements:

90 degree Scale, 6 Inch Rectangular Switchboard
Frequency Meters



ITEM	REQUIREMENT
1. Scale length	5 inches min.
2. Normal operation position	Mounted on vertical panel.
3. Position influence	60-degree tilt from normal operating position, 1 percent of scale length max. with rated voltage applied.
4. Rated accuracy	3.0 percent.
5. *Damping factor*	1.5 *min.* (*overshoot*, 67-percent max.)
6. Response time	5 seconds max. at rated voltage.
7. Loss	5 va max. at middle frequency with rated voltage applied.
8. External temperature influence	1 percent of frequency span max. at middle frequency with ±10°C. variation from +25°C. with rated voltage applied.
9. External field influence	4 percent of frequency span max. at middle frequency in 5 oersted field with rated voltage applied.
10. Safe working voltage to ground	1100 volts max., peak value.
11. Dielectric test	Live parts to face and panel: 2600 volts a.c.
12. Max. voltage	20 percent greater than rated value.
13. Voltage influence	0.5 percent of frequency span with ±10 percent voltage variation; 4 times this max. error is allowed for a ±20 percent variation.
14. Temperature, permanent effect	2 percent of frequency span.
15. Vibration, permanent effect	2 percent of frequency span.
16. Humidity resistance	3 percent of frequency span.

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:20Z
Check the source to verify that this is the current version before use.

3.24.4 Detail requirements for portable frequency meter, 1.5 percent accuracy class:

1. Scale length	5 inches min.
2. Position influence	1 percent of scale range, max. change.
3. Initial accuracy	1.5 percent max. error.
4. Damping factor	3 min.
5. Response time	3.5 seconds max.
6. Voltage influence	0.75 percent of frequency span with ± 10 percent voltage variation.
7. Loss	10 va max. at middle frequency with rated voltage applied.
8. External temperature influence	One percent of frequency span max. at middle frequency with ±10°C. variation from +25°C. with rated voltage applied.
9. External field influence	2 percent of frequency span at middle frequency in 5 oersted field with rated voltage applied.
10. Heat effect at 65°C.	1.0 percent max. change at 65°C. 0.5 percent max. permanent effect.
11. Low temperature exposure, 0°C.	1.0 percent max. change at 0°C. 0.5 percent max. permanent change.
12. Humidity resistance	2 percent max. change.
13. Vibration	1 percent max. change.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility of inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification, test conditions and general instructions. - The methods of sampling, inspection, and test conducted on meters shall be classified as follows:

(a) Qualification tests (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.2.1 Atmospheric conditions. -

4.2.1.1 Qualification tests. - Unless otherwise specified herein, all qualification tests shall be made at ambient temperature of 23° to 27°C., at existing barometric pressure, and with relative humidity of 40 to 60 percent.

4.2.1.2 Quality conformance tests. - Unless otherwise specified herein, all tests shall be made at ambient temperature of 20° to 30°C., at existing barometric pressure, and with relative humidity less than 80 percent.

4.2.2 Reference temperature. - The reference temperature shall be 25°C.

4.2.3 Position. - Unless otherwise specified herein, tests performed on meters, including all tests requiring scale readings, shall be performed with the meter in the position in which it is normally used.

4.2.4 Examination of pivots and bearings. - Examination of pivots and bearings for evidence of corrosion, damage from vibration, shock, and other tests, shall be made through a microscope having a magnification of 40 to 100 diameters.

4.2.5 Tapping. - When taking meter readings, either the meter or its support shall be tapped lightly, except when ascertaining the effect of friction after the vibration test.

4.2.6 Parallax. - Care should be exercised to avoid the effect of parallax when taking meter readings.

4.2.7 Resetting to zero. - Unless otherwise specified herein, meters provided with an external zero-adjuster may be set to zero at the completion of each test.

4.2.8 Thermal stability. - For all tests, care shall be taken to obtain thermal stability at the specified temperature prior to reading the meter.

4.2.9 Number of scale readings. -

4.2.9.1 Switchboard meters. - Unless otherwise specified herein, readings on switchboard

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:20Z
Check the source to verify that this is the current version before use.

meters shall be taken at six approximately equidistant points on the scale including the zero and end-scale readings.

4.2.9.2 Portable meters. - On single-range portable meters, readings shall be taken at each numbered cardinal scale mark.

4.2.10 Standard meters. - Standard meters shall be used to check the indications of meters submitted for test. Accuracies and tolerances specified are absolute, and readings of standard meters should be corrected before determining compliance.

4.3 Qualification tests[1/]. - Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified herein.

4.3.1 Samples. - Sample meters submitted for qualification approval shall be representative of the manufacturer's normal production and shall be taken, preferably, from a current production lot. The type designation and number of specimens required to be submitted for each kind of meter shall be as specified in table I, as applicable. Where 400 cycle meters are similar in basic construction to the submitted 60-cycle meter, only one additional specimen of each type is required. All qualification tests shall be performed on this specimen except the tests for external-field influence, vibration, humidity, resistance, and shock (high-impact). If the basic design is different, a full number of specimens shall be submitted, and all qualification tests shall be performed.

Table I - Samples for qualification tests.

Kind of meter	Range in cycles	Number of specimens required
4-1/2 inch, 250° scale	59-61	3
	55-65	3
Class HI shockproof	390-410	3
Switchboard meter	350-450	3
4-1/2 inch, 250° scale	59-61	2
	55-65	2
Switchboard meter	390-410	2
	350-450	2
4-1/2 inch, 90° scale	55-65	2
Switchboard meter	350-450	2

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3 and 6.4).

Table I - Samples for qualification tests (cont'd)

Kind of meter	Range in cycles	Number of specimens required
6 inch, 90° scale	55-65	2
Switchboard meter	350-450	2
Portable meter	55-65	1
	350-450	1

4.3.1.1 Test routine. - Sample meters shall be subjected to the applicable qualification tests shown in table II in the order listed.

Table II - Qualification tests.

Test	Requirement paragraph	Test paragraph
Visual and mechanical inspection (without opening case)	(See table IV)	4.4
Zero adjuster	3.5.3.2	4.3.2
Position influence (balance)	3.7	4.3.3
Initial accuracy	3.8	4.3.4
Damping factor	3.9	4.3.5
Response time	3.10	4.3.6
Power consumption	3.11	4.3.7
External-field influence	3.13	4.3.8
Heat effect at 65°C.	3.14	4.3.9
Temperature influence	3.15	4.3.10
Low-temperature exposure	3.16	4.3.11
Dielectric stress	3.17	4.3.12
Insulation resistance	3.18	4.3.13
Vibration	3.19	4.3.14
Humidity resistance (1 specimen)	3.20	4.3.15
Shock (high-impact) (2 specimens)	3.21	4.3.16
Visual and mechanical inspection (external and internal)	(See table IV.)	4.4

4.3.2 Zero adjustment. - The range of the zero adjustment above and below the center of the zero mark shall be determined. Record shall be made as to whether or not the zero-adjuster may be rotated through 360° without damage to the meter. (See 3.5.3.2.)

4.3.3 Position influence (balance). - The pointer shall be set to the center of the zero scale division with the meter in the normal operating position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:20Z
Check the source to verify that this is the current version before use.

The meter shall then be tilted 60° from this position in any direction, and the deviation of the pointer noted. (See 3.7.3.)

4.3.4 Initial accuracy. – Meters shall be checked for accuracy as specified in 4.3.4.1 or 4.3.4.2, as applicable.

4.3.4.1 Qualification. – After preliminary conditioning in the test circuit for 30 minutes at center scale deflection, readings shall be taken and compared with corresponding readings of the standard meter. The test shall be made with the meter at a temperature of 25°C. (See 3.8 and 4.2.9.)

4.3.4.2 Quality conformance. – The quality conformance test of initial accuracy shall be performed under the test conditions specified in 4.2 without preliminary conditioning. (See 3.8.)

4.3.5 Damping factor. – Electric power sufficient to produce momentary end-scale deflection shall be applied to the meter. The maximum momentary deflection shall be observed and recorded. After the pointer has come to rest, the steady deflection shall be observed and recorded. The damping factor shall be taken as the ratio of the steady deflection (in angular units) to the difference between the maximum momentary deflection and the steady deflection in angular units. Example: If the maximum momentary deflection is 90 angular degrees and the steady deflection is 75 angular degrees, the difference between the two is 15°, and the damping factor (the ratio of 75° to 15°) is 5. (See 3.9.)

4.3.6 Response time. – Steady electric power sufficient to produce momentary end-scale deflection shall be applied to the meter. The length of time, in seconds, required for the pointer to come to apparent rest shall be recorded. The pointer shall be considered as having come to apparent rest when it has obtained the actual point of rest within one-half the initial-accuracy requirement. The test shall be repeated five times, and the average length of time recorded for all measurements shall be considered as the response time of the meter. (See 3.10.)

4.3.7 Power consumption. – The power consumption shall be measured at end-scale deflection and recorded in terms of current, voltage, volt-amperes, or ohms, as applicable. (See 3.11.)

4.3.8 External-field influence. – Meters shall be placed at the center of a four-turn coil, 50 centimeters in diameter, carrying a current of 50 amperes of the same kind and frequency as that which actuates the meter mechanism, and which produces a field of approximately 5 oersteds at the center of the coil. Readings shall be taken with the meter at various positions in the center of the field. (See 3.13.)

Note. – It is the intent to specify a test coil which will produce a uniform field of approximately 5 oersteds through the case of the meter. Therefore, coils of other sizes are equally suitable if an equivalent electrical field is produced.

4.3.9 Heat effect at 65°C. – Meters shall be maintained at reference temperature for a sufficient time (not less than 2 hours) to attain thermal stability, and shall then be subjected to the three successive cycles of temperature change listed in table III. During this test, meters shall indicate freely (see 3.2.9) at all parts of the scale. (See 3.14.)

Table III - Procedure for determining heat effect at 65°C.

Cycle number	Period (hours)	Step number	Procedure
		1	Take initial reading at reference temperature.
	16	2	Place meter in chamber at 65°C. and maintain that temperature for the remainder of that period.
		3	Read meter while at 65°C. at the end of step 2.
	8	4	Allow chamber and meter to cool down to the reference temperature during this period.
1		5	Read meter while at reference temperature at the end of step 4.
	16	6	Raise temperature of chamber and meter as rapidly as possible to 65°C. and maintain that temperature for the remainder of the period.

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:20Z
Check the source to verify that this is the current version before use.

Table III - Procedure for determining heat effect at 65°C—(Continued)

Cycle number	Period (hours)	Step number	Procedure
2		7	Read meter while at 65°C. at the end of step 6.
	8	8	Allow chamber and meter to cool down to the reference temperature during this period.
		9	Read meter while at reference temperature at the end of step 8.
	16	10	Raise temperature of chamber and meter as rapidly as possible to 65°C. and maintain that temperature for the remainder of the period.
		11	Read meter while at 65°C. at the end of step 10.
3	8	12	Allow chamber and meter to cool down to the reference temperature during this period.
		13	Read meter while at reference temperature at the end of step 12.

4.3.10 Temperature influence. – Meters shall be tested at temperature 10°C. above and below the reference temperature. Readings shall be taken at not less than four points on the scale, one of which is at or near the maximum scale value, and the temperature influence computed. The temperature influence shall be considered one-half the percentage change in indication caused by the 10°C. from the reference temperature. If the influences above and below the reference temperature are not equal, the greater value shall be considered the temperature influence. (See 3.15.)

4.3.11 Low-temperature exposure. – Meters shall be exposed to a temperature of 0°C. for 16 hours, and at the end of the period meter readings shall be taken. It shall be observed whether or not the meter indicated freely at all parts of the scale while at 0°C. Meters shall then be maintained at room temperature for sufficient time to attain thermal stability, and meter readings repeated. The permanent change in indication resulting from the low-temperature exposure shall be noted. (See 3.16.)

4.3.12 Dielectric stress. – When performing this test, portable meters with nonmetallic cases shall be placed on a metallic platform; switchboard meters with nonmetallic cases shall be mounted on a metallic panel. All meters shall be subjected to a voltage of 2,600 volts r.m.s. applied between all external live parts of the meter and:

(a) The metallic platform or panel on which the meter is placed or mounted, or the meter case, if the case is metallic.

(b) The zero-adjuster; and range selector switch, when applicable.

(c) All exposed metal parts not a live part of the meter.

The voltage shall be nominal 60 cycles per second in frequency and shall aproximate, as closely as possible, a true sine wave in form. The voltage shall be applied continuously for a period of 60 seconds. The test voltage shall be measured by the voltmeter method whereby the measuring meter derives its voltage from the high-potential circuit, either directly or by means of a voltmeter coil placed in the testing transformer, or by means of an auxiliary ratio transformer. To avoid voltage surges, the applied voltage shall be raised to its full value gradually, and upon completion of the 60-second test period shall be gradually reduced. The test shall be made using high-potential test prods having not less than 1/8-inch radius tips. Any damage or flash-over shall be noted. (See 3.17 and MIL-STD 202.)

4.3.13 Insulation resistance. – Meters shall be placed on a metallic platform or mounted on a metallic panel as specified in 4.3.12. Using a potential of 500 volts direct current, the insulation resistance shall be determined between all external live parts and the metallic platform or panel on which the meter is placed or mounted, or the meter case, if the case is metallic. (See 3.18 and MIL-STD 202.)

Note. – For quality conformance tests of meters furnished with external accessories as an integral part, the complete meter (meter with accessories) shall be tested.

Source: https://assist.dla.mil -- Downloaded: 202640-18T08:20Z
Check the source to verify that this is the current version before use.

4.3.14 Vibration. - Vibration tests shall be conducted on meters as follows:

(a) Class HI shockproof meters. - Class HI shockproof meters shall be tested for vibration in accordance with type I of MIL-STD-167.
(b) Other meters. - Meters shall be mounted in various positions upon a vibration platform and subjected to vibration for a total period of 24 hours. The construction of the vibration platform shall be such as to give simple harmonic motion at a frequency of 400 cycles per minute, and with a maximum amplitude of one-sixteenth inch (one-eighth inch maximum total excursion) at the horizontal centerline of the meter. The change in indication shall then be determined. (See 3.19)

4.3.14.1 Friction effect. - Upon completion of the vibration test, the friction effect shall be determined, as follows: Untapped ascending and descending readings shall be taken at five points distributed over the scale, using the same scale points for ascending and descending readings. The difference between corresponding ascending and descending readings as observed on the standard meter shall be determined. (See 3.19.1.)

4.3.15 Humidity resistance. - Meter readings shall be taken at reference temperature. The meter shall then be placed in a humidity chamber and kept for 6 hours at a temperature of 65°C. with relative humidity of 90 percent. Sufficient electrical energy shall be supplied to the meter to maintain two-thirds end-scale deflection. At the end of the 6-hour period, the heat shall be shut off and the meter de-energized. The meter shall then be permitted to cool for 18 hours in this atmosphere, the relative humidity of which rises as the temperature is reduced. (Precaution shall be taken to prevent dripping on the meter.) A temperature of 35°C. or lower, but not less than 25°C., shall be attained during the cooling period. The complete cycle shall then be repeated. Upon completion of the second cycle, the meter shall be removed from the humidity chamber and permitted to dry for a period of 24 hours at room temperature, with relative humidity less than 60 percent. Meter readings shall then be taken at reference temperature. The meter shall then be removed from the humidity chamber and permitted to dry for a period of 24 hours at room temperature, with relative humidity less than 60 percent. Meter readings shall then be taken at reference temperature. After the drying period, a maximum of 10 momentary end-scale deflections will be permitted prior to the final reading. Only one meter of the sample shall be subject to this test. (See 3.20 and MIL-STD 202.)

4.3.16 Shock (high-impact). - High-impact-shock-resistant meters shall be tested in accordance with type C of MIL-S-901 for lightweight equipment, when mounted on a plate as shown on figure 6D of that specification. Two meters of the sample, which were not subjected to the humidity-resistance test, shall be used. During the test, one of the meters shall be energized to approximately one-half scale deflection. A total of nine blows shall be applied. Three blows shall be applied parallel to each of the three principal axes of the meters, the three blows for each direction to be with heights of hammer drop 1 foot, 3 feet, and 5 feet. The pointer shall be reset to zero after each blow. At the completion of the shock test, the meters shall be checked for calibration errors, then disassembled and carefully examined for damage or misalinement of parts. (See 3.21 and MIL-STD 202.)

4.4 Sampling procedure. - At the discretion of the Government, quality conformance tests may be conducted by either of the following methods or a combination of both:

(a) 100-percent inspection.
(b) Statistical sampling and inspection in accordance with MIL-STD-105. The acceptable quality level (AQL) shall be as specified in table IV. The classification of major and minor defects for visual and mechanical inspections shall be as specified in table V.

4.4.1 Test routine. —Meters shall be subjected to the tests specified in table IV, in the order listed.

Table IV. - Quality conformance inspection.

Test	Requirement paragraph	Test paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical inspection (without opening case)	(See table V)	4.4.3	1.5	4.0
Position influence (balance)	3.7	4.3.3		
Initial accuracy	3.8	4.3.4.2		
Dielectric stress	3.17	4.3.12	1.5	
Damping factor	3.9	4.3.5		

4.4.2 Rejected meters. - Rejected meters shall be returned to the contractor. Rejected meters may be reworked, if possible, in such a manner as to remove all defects. Reworked meters shall be submitted for acceptance separately from newly-produced meters.

Source: https://assist.dla.mil -- Downloaded: 20260-18T08:20Z
Check the source to verify that this is the current version before use.

4.4.3 Visual and mechanical inspection. - Meters shall be inspected to determine compliance with table V.

Table V. - Visual and mechanical inspection.

Requirement	Paragraph	Defect classification
Material	3.3 to 3.3.4, incl.	Major
Design and construction	3.4	Minor
Cases	3.5.1 to 3.5.1.4.1.2, incl.	Major
Dial and pointer	3.5.2	Minor
Means of adjustment	3.5.3	Minor
Terminals and connections	3.5.4	Minor
Mounting switchboard meters	3.5.5 to 3.5.5.2, incl.	Major
Marking	3.22	Minor
Workmanship	3.23 and 3.23.1	Major

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.3 Marking. - Shipment information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include *nomenclature*, Federal stock number or manufacturer's part number, type, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment requirements. - The following provides various levels of protection during domestic shipment and storage or overseas shipment which may be required when procurement is made by a Government activity. (See 6.2).

5.2.1 Preservation, packaging and packing. - The equipment shall be preserved and packaged by levels A or C and packed by levels A or B in accordance with MIL-E-17555.

5.2.2 Marking. - In addition to any special marking required, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. -

6.1.1 Switchboard meters. - Switchboard meters are intended for use in general application of electrical measurement including shipboard and land installations. High-impack-shock-resistant meters are preferred for shipboard use.

6.1.2 Portable meters. - Portable meters are suitable for shipboard use as primary and secondary standards; however, they are intended for use in other applications in which the specified accuracy is sufficient for the required measurements. More accurate meters, generally designed as laboratory standards, are not covered by this specification.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Complete type designation, and title, number, and applicable detail requirement. (See 1.2.1 and 3.24.)
(c) External accessories, if required. (See 3.6.)
(d) Preservation, packaging, packing or marking requirements other than those required by paragraph 5.1. (See 5.2.)

6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 16125, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (*see 6.4*).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:20Z
Check the source to verify that this is the current version before use.

6.4 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pa.

6.5 Precautions. - Attention of equipment designers and manufacturers is invited to the fact that sufficient space should be provided in equipment for mounting standard meters of the maximum depth specified (see 3.24) even though a shallower meter may be in the initial design.

6.5.1 Red mark on dial. - When specified in the contract or order, the manufacturer may place red marks on meter dials at designated positions. There will be no change in the type designation to distinguish such meters from those having normal markings; consequently, a meter supplied for a certain application may have a red mark at an undesired position. If this occurs, it is recommended that the user remove or cover the red mark by any convenient method, and make other red marks, if desired.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:
Army - EL
Navy - Ships
Air Force - 2750

Preparing activity:
Navy - Ships
(Project 6625 - 0081)

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:20Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Sea System Command (SEA 3112)
Department of the Navy
Washington, DC 20362

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:20Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil - Downloaded: 2026-10-18T08:20Z
Check the source to verify that this is the current version before use.